

FOUNDATIONS OF THE RELATIONAL HOLONIC CALCULUS

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ABSTRACT. This working monograph reconstructs the most durable mathematical content of the Soma research programme as a dependency network rather than a chronology of claims. Its core is an occurrence-level calculus in which actual current meets available Standing in one atomic event, producing one successor and immediate radiation; a genuinely later consequence can then change later conduct. The manuscript derives receiver-relative comparison, completion, and exact compression; develops finite exponential, Euler-product, theta/Mellin, and aperture-tail relations without claiming analytic completion or the Riemann hypothesis; types the links to hypergraph rewriting, manifolds, fractal return, physics, and living systems; distinguishes a learned relation from one neural factorization; and states the measured boundary of the production machine. Direct constraints, assistant derivations, external theorems, implementation facts, and measurements remain explicitly distinct.

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1. ORIENTATION

CONTENTS OF THIS PART

Five kinds of sentence

The research object

This volume replaces a prospectus which catalogued possible papers but did not yet expose their mathematical dependence. The present order follows the requested research sequence: foundations (J1), exact formula and RH-facing mathematics (J4), comparative geometry and physics (J5), learning and modern machine learning (J3), and machine realization (J2). The order is epistemic, not chronological. A later measurement may depend on an earlier definition; an older external theorem may enter only after the receiver which uses it has been declared.

The central methodological rule is simple: a shared word or visual resemblance never supplies a mathematical identification. A relation crosses domains only through an explicit map whose assumptions, receiver, and failure modes are stated. Thus a triangle may be an oriented boundary, a parallel-path comparison, a local rewrite hinge, or a geometric simplex. These are compatible readings only when the relevant structures have actually been supplied.

AUTHORSHIP AND STANDING. Brandon's direct constraints, hypotheses, intuitions, questions, and corrections open the research direction. Definitions, derivations, reconstructions, and proposed formal consequences written by Sol remain assistant contributions unless independently established. External theorems retain their original authorship. Machine claims are restricted to the named implementation or observation.

Source route. The conversation opens the hypothesis; the record preserves who said what .

Open boundary. Ratification preserves a presented relation at its stated boundary. It does not convert an analogy into identity, a modeling choice into theorem, or a predicted signature into a measurement.

1.1. Five kinds of sentence.

The manuscript uses five compact grades. A **direct constraint or hypothesis** records Brandon's stated boundary or research posture. A **working definition** fixes terminology for this calculus. An **exact derivation** follows from written definitions or inherited mathematics. An **implementation or measurement** reports source or runtime evidence at a named cut. An **open bridge** names the mechanism, theorem, or observation still missing. These grades are orthogonal: an exact derivation need not be physically realized, and an exact runtime observation need not generalize.

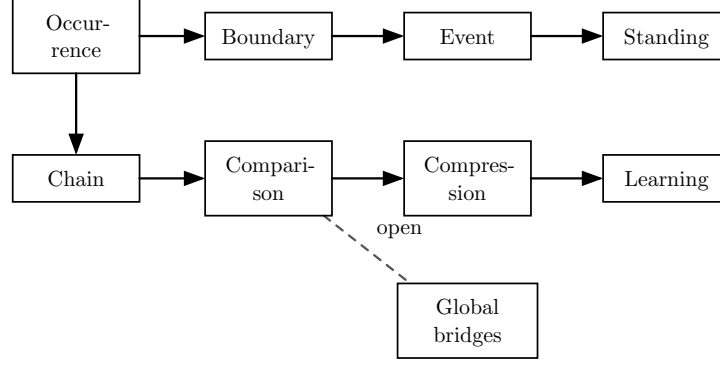


FIGURE 1. Dependency graph of the foundation. Solid arrows are definitions or exact local constructions developed here. The dashed arrow collects distinct analytic and physical obligations which are not supplied by the calculus.

The solid portion of Figure 1 is the theory developed in this volume. The rightmost dashed edge is intentionally not a single conjecture. In the RH direction it contains analytic continuation, a complete explicit-formula carrier, and positivity on an admissible test space. In the physical direction it contains domain laws, units, covariance, limiting correspondence, and discriminating measurement. Holonics supplies a language for keeping these obligations connected without pretending that they have already been discharged.

1.2. The research object.

Definition 1.2.1. A **relational ecology at a receiver** is a tuple

$$\mathcal{E}_\rho = (O, I, D, T; K_t, S_t, F_\rho),$$

where O is a population of actual occurrences, I is oriented boundary incidence, D is causal dependency, T is local transport, K_t is current material, S_t is available Standing, and F_ρ is the face available to receiver ρ . No component is inferred merely from equality of labels, coordinates, values, or rendered positions.

The word **holon** will therefore not name a permanently bounded object with intrinsic scale. It names a receiver-relative consequential relation which may continue through changing constituents. “Whole” and “part” are faces of a live construction, not absolute types attached to atoms.

SOURCE ROUTE. Routine claim routing comes from the Soma claim and evidence index. The formal spine is reconstructed from Elementary Mechanics, the Formula, and the Relational Holonic Calculus deposit. These sources are evidence and provenance, not a second sequence of definitions beside the one fixed here.

2. J1. RELATIONAL HOLONIC FOUNDATIONS

CONTENTS OF THIS PART

Actual occurrence precedes algebra	Four structures, one optional coupling
Judgments and lawful forgetfulness	The atomic event
Comparison: RIDE, FOUND, and OPEN	Completion and higher grain
Exact compression is relative to future questions	Learning as a causal contrast

The earlier attempts at holonics often asked one representation - a manifold, hypergraph, triangle, circuit, or field - to carry every aspect of the theory. The rigorous correction is to begin with several typed relations and connect them only where a construction supplies the connection. Occurrence is not boundary; boundary is not dependency; dependency is not transport; transport is not a metric; and a receiver face is not the whole construction.

2.1. Actual occurrence precedes algebra.

Definition 2.1.1. An **occurrence carrier** is a graded population

$$O = \cup_{k \geq 0} O_k$$

of actual cells, with the grades declared pairwise disjoint. Two elements of O remain distinct when they have equal values or equal boundary inscriptions but arose as distinct incidences. An oriented incidence relation assigns to each $\sigma \in O_k$ a finite signed boundary

$$\partial\sigma = \sum_{\tau \in O_{k-1}} \varepsilon_{\sigma,\tau} \tau, \quad \varepsilon_{\sigma,\tau} \in \mathbb{Z}.$$

The population language prevents a recurring error: a coefficient cancellation is not the annihilation of the cells which contributed the coefficients. If two triangles share one edge, their opposed orientations may cancel on that **same occurrence** after gluing. Equal but unglued edge labels do not cancel.

Proposition 2.1.2. Let $C_k = \mathbb{Z}[O_k]$ be the free abelian group on actual k -occurrences. If the supplied incidence obeys

$$\sum_{\tau} \varepsilon_{\sigma,\tau} \varepsilon_{\tau,v} = 0$$

for every $\sigma \in O_k$ and $v \in O_{k-2}$, then the linear extension $\partial_k : C_k \rightarrow C_{k-1}$ satisfies $\partial_{k-1} \partial_k = 0$.

Proof. It is enough to evaluate a basis occurrence σ . The coefficient of v in $\partial\partial\sigma$ is exactly $\sum_{\tau} \varepsilon_{\sigma,\tau} \varepsilon_{\tau,v}$, which vanishes by the incidence assumption. Linearity extends the result to every chain. $\square$

This proposition is deliberately modest. It establishes the chain-complex face of supplied incidence. It does not infer a rewrite rule, metric, causal dependency, or probability distribution. Conversely, a local rewrite may be well-defined even when the cells have dimension zero; dependency rank, topological dimension, and constituent grain are independent coordinates.

SOURCE ROUTE. The occurrence-level correction and its implementation requirements are stated in Elementary Mechanics, §§2-4 and measured in Parameterized elementary mechanics 01 . The mathematical chain boundary follows the standard construction in [1].

2.2. Four structures, one optional coupling.

For a local region U , the minimal carrier is

$$\mathcal{K}_U = (O_U, I_U, D_U, T_U).$$

Here I_U is oriented boundary incidence; D_U is actual production or preservation dependency; and T_U assigns a carrier map to an admitted directed incidence. For a path $\gamma = a_n \dots a_1$,

$$T_\gamma = T_{a_n} \circ \dots \circ T_{a_1}.$$

A hypergraph may supply arbitrary-arity local relations and a rule may replace such relations, but neither supplies orientation or transport on its own. A chain complex supplies oriented cancellation, but not which replacement is afforded. A connection or functorial transport supplies path comparison, but not which path occurred. This four-way separation is the smallest stable foundation found in the record.

Definition 2.2.3. A **preserved rewrite interface** is a span

$$L \xleftarrow{\ell} K \xrightarrow{r} R$$

in a declared category of local structures. The rewrite replaces the image of L by the image of R while retaining the actual occurrence-level image of K . A rendered coordinate, repeated label, or equal value is not an interface unless the source supplied the corresponding maps.

Double-pushout rewriting in an adhesive setting gives one rigorous realization of this definition [2]. It is a specialization, not a primitive demanded of every Eros world. Likewise Pachner moves are lawful retriangulations of piecewise-linear manifolds under their own hypotheses [3]; they do not define general learning or compression.

2.3. Judgments and lawful forgetfulness.

The calculus needs more than a single equality symbol. We use the following judgments:

Notation	Meaning and jurisdiction
$\Gamma \vdash A = B : T$	construction identity in declared type T
$\Gamma \vdash A \underset{\tau}{\simeq} B$	reversible gauge or chart transport τ
$\Gamma \vdash A \underset{\rho_K^{F,g}}{\equiv} B$	agreement at receiver ρ , frame F , and grain g
$\Gamma \vdash A \rightarrow B$	directed construction or worldline under current K
$\Gamma \vdash A \overset{q}{\Rightarrow} Q$	declared compression or quotient q

TABLE 1. Core notation. Every symbol is indexed by a declared receiver, frame, or boundary when that index can change the result.

Construction identity is the strongest judgment. Gauge equivalence requires explicit reversible transport. Face agreement states only that a named observation cannot distinguish the two inputs. A worldline is directed and need not be invertible. A compression may intentionally discard distinctions which no declared later contact uses.

Lemma 2.3.4. Let $q : X \rightarrow Q$ be a non-injective observation quotient. There is no map $i : Q \rightarrow X$ satisfying $i \circ q = \text{id}_X$.

Proof. Choose $x \neq y$ with $q(x) = q(y)$. If such i existed, then $x = i(q(x)) = i(q(y)) = y$, a contradiction. $\square$

The lemma is the elementary obstruction behind several project rules. An endpoint does not reconstruct a path, a token does not reconstruct its full source interval, an observed output does not identify a network interior, and a compressed successor does not preserve undeclared future distinctions unless a lift or congruence theorem is supplied.

2.4. The atomic event.

Definition 2.4.5. One event at time cut t is a partial receiving map

$$\Lambda_{F_e, Q_e} : (S_t, K_e) \rightarrow (S_{t^+}, R_e),$$

where S_t is Standing-before, K_e is the complete co-present current admitted to event e , F_e is the source-declared receiver or world structure, Q_e is the objective or exposed boundary when present, S_{t^+} is one successor, and R_e is immediate radiation. Event testimony π_e can be formed only after commit.

Axiom 2.4.6. Same-prestate atomicity. Every co-present constituent of K_e meets the same unchanged S_t . Validation, integration, and commitment produce either one S_{t^+} or no successor. An executor may perform physical conduct inside the call, but it cannot own or reconstruct the successor from a receipt.

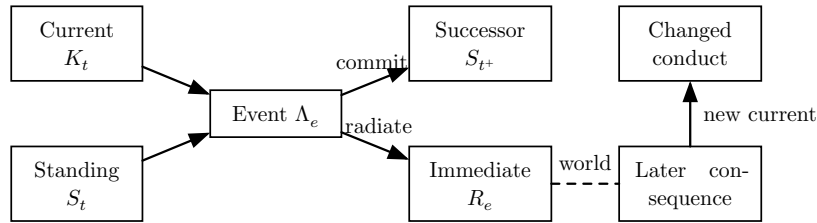


FIGURE 2. One event and its later causal circuit. Immediate radiation and a genuinely later world consequence are different cuts.

Several invariants follow immediately from the event boundary:

- no arbitrary filesystem, caller, or hardware order may manufacture chronology or incidence;
- immediate radiation is not a later consequence;
- exact rest performs no event;
- one committed successor has one visibility edge;

- attribution records what the source supplied and what the meeting formed; and
- co-present currents cannot be serialized into false causal instants merely to fit an implementation.

World, Current, Standing, and Lineage are not four substances. They are projections of one live relation at a selected cut. A constituent may leave active participation while its consequential relation continues at a higher grain. Lineage is therefore continuity of causal participation, not storage of a perfect ancestry copy.

SOURCE ROUTE. The current authority is Formula §§C, CIV-CV together with The event emanates the successor; the executor cannot reconstruct the machine . The latter records the source correction which removed receipt-owned successor reconstruction.

2.5. Comparison: RIDE, FOUND, and OPEN.

For the oriented triangle

$$x_0 \xrightarrow{a_{01}} x_1 \xrightarrow{a_{12}} x_2, \quad x_0 \xrightarrow{a_{02}} x_2,$$

the boundary is

$$\partial\tau = a_{12} - a_{02} + a_{01}.$$

Transport supplies the complete path pair

$$X_\tau = (T_{a_{12}} \circ T_{a_{01}}, T_{a_{02}}).$$

If the codomain has an additive chart, the optional residual is the difference of these routes. Without such a chart, the ordered pair itself is the complete comparison.

Definition 2.5.7. Relative to receiver ρ and grain g , a comparison **RIDEs** when prior supported transport conducts through the actual meeting; it **FOUNDs** when the event actually establishes new support; and it remains **OPEN** when the compared routes or boundary do not close at that cut. Noncommutation is OPEN by default. It is not falsehood, automatic branching, or a permanent prohibition on later contact.

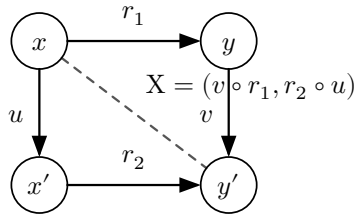


FIGURE 3. A complete path comparison. Equality may close the square; otherwise both routes remain material and the comparison is OPEN at this receiver.

Proposition 2.5.8. If the square in Figure 3 is evaluated in a category with the displayed composites, then commutation is the equality

$$v \circ r_1 = r_2 \circ u.$$

Failure of this equality determines a complete comparison pair but, without additional structure, determines neither a scalar residual nor a new filler.

Proof. The two composites are well-typed parallel arrows. Their equality is therefore meaningful. A subtraction requires an additive hom-set; a quotient requires invertibility or a chosen projective structure; and a filler requires an existence law or actual new event. None follows from the mere fact that the arrows are unequal. $\square$

This is the rigorous content behind the project's use of OPEN as useful terrain. A later actual boundary may match and RIDE while the earlier comparison remains truthfully OPEN at its earlier scope.

2.6. Completion and higher grain.

Definition 2.6.9. Let $B \subseteq O_{k-1}$ be the declared exterior boundary family. A finite k -chain c is **complete relative to B** when

$$\partial c \in \mathbb{Z}[B].$$

A **hand-up** additionally requires an actual event which exposes the resulting boundary as one higher-grain occurrence. Algebraic closure alone does not manufacture that occurrence.

Proposition 2.6.10. Suppose c_1 and c_2 are relatively complete with respect to B , and their shared interior faces are the same occurrence with opposed coefficients. Then $c_1 + c_2$ is relatively complete with respect to B , and the shared face is absent from its boundary.

Proof. Linearity gives $\partial(c_1 + c_2) = \partial c_1 + \partial c_2$. The opposed coefficients on the shared basis occurrence cancel. All remaining terms lie in $\mathbb{Z}[B]$ by hypothesis. $\square$

The proposition explains composition, not deletion. The two incident cells and their causal contribution may remain in the higher constituent even though the shared boundary is no longer exposed.

2.7. Exact compression is relative to future questions.

Definition 2.7.11. Let $\Phi = \{\varphi_u : X \rightarrow Y_u\}_{u \in U}$ be a declared observation family. A quotient $q : X \rightarrow Q$ is **exact for Φ** when for every u there exists $|\varphi|_u : Q \rightarrow Y_u$ such that

$$\varphi_u = |\varphi|_u \circ q.$$

Theorem 2.7.12. Let X, Q, Y_u be modules and all maps be linear. The quotient $q : X \rightarrow Q$ is exact for Φ if and only if

$$\ker q \subseteq \{x : \varphi_{u(x)} = 0 \text{ for every } u \in U\},$$

provided q is surjective.

Proof. If $\varphi_u = |\varphi|_u \circ q$, then $q(x) = 0$ implies $\varphi_{u(x)} = 0$, proving the inclusion. Conversely, define $|\varphi|_{u(q(x))} = \varphi_{u(x)}$. If $q(x) = q(x')$, then $x - x' \in \ker q \subseteq \ker \varphi_u$, so the definition is independent of representative. Surjectivity defines $|\varphi|_u$ on all of Q . $\square$

Exactness for a present observation family does not imply exactness for every later world. The stronger condition is a future congruence: whenever $q(x) = q(y)$, every admitted future current must continue x and y to states whose relevant faces still

agree. The production implementation therefore validates outgoing compression against the complete exposed later-contact boundary of the actual event, not against an imagined universal observer.

2.8. Learning as a causal contrast.

Definition 2.8.13. Let e be an earlier event and p a genuinely later probe. Learning at receiver ρ is established when

$$\text{Conduct}_{\rho(S_{\text{after } e}, p)} \neq \text{Conduct}_{\rho(S_{\text{without } e}, p)}$$

and there is an admitted participation path from the relation formed by e to the changed later conduct. Mere co-occurrence, testimony, or a source-provided answer does not satisfy the criterion.

The counterfactual is local. It may compare a no-Standing sibling, original Standing before cultivation, a wrong interface, or a retained exact foil. It does not demand that learning occur without inherited identities, tokenizers, models, arithmetic, or culture. Those are lawful Standing when their attribution is retained.

Corollary 2.8.14. Repetition without participation does not establish learning, while source-departed Standing can establish learning if it changes the later probe relative to a matched foil.

This completes the foundational chain:

occurrence $\rightarrow$ incidence $\rightarrow$ event $\rightarrow$ successor $\rightarrow$ Standing $\rightarrow$ later contrast.

Geometry, probability, optimization, and physics enter only as optional structures on this chain. The next part shows how far exact formula ecology can proceed under that discipline.

EVIDENCE RECEIPT. Claim. The foundational calculus supplies typed local derivations and a testable causal criterion.

Grade. Working definitions plus exact algebraic consequences; selected mechanisms are implemented and measured later in J2.

Support. Elementary Mechanics; Formula §§C, CIV-CV, CXLVI, CL-CLI; current README evidence routes.

Open boundary. No complete categorical semantics, reference checker, general confluence theorem, universal observation family, or automatic source-interface discovery is claimed.

3. J4. FINITE GENERATORS, PRIMES, AND THE RH BOUNDARY

CONTENTS OF THIS PART

Formal exponential at finite degree	Prime powers are required by the Euler exponential
Aperture growth without replay	Two exact atlases for pi
Scale and turn meet on the logarithmic cover	Finite theta involution
The residual is a moving boundary	The exact RH boundary

The mathematical programme around π , e , primes, and the Riemann hypothesis becomes coherent once three questions are separated:

1. Which finite object generates the next exact face?
2. Which equality is a formal or convergent theorem in its declared domain?
3. Which analytic completion and positivity statement is still absent?

The finite object is not called infinite merely because it presents the first terms of an unbounded family. It carries a seed, a next-face law, an admissible boundary, an exact residual or enclosure, and a rebase from one aperture to the next.

3.1. Formal exponential at finite degree.

Let $A = \prod_{n \geq 0} A_n$ be a completed graded commutative rational algebra and let $A_+ = \prod_{n \geq 1} A_n$ be its augmentation ideal. For $F \in A_+$ define

$$\exp(F) = \sum_{k \geq 0} \frac{F^k}{k!}.$$

At total degree at most N , only $k \leq N$ can contribute, so the restriction is finite. A nonzero degree-zero term would require another topology or analytic notion of convergence and is not silently admitted.

Proposition 3.1.1. In the quotient $\frac{\mathbb{Q}[y]}{y^{N+1}}$,

$$\exp(y) = \sum_{n=0}^N \frac{y^n}{n!}.$$

Evaluation at $y = 1$ gives the exact rational aperture

$$E_N = \sum_{n=0}^N \frac{1}{n!}.$$

Proof. Every term y^n with $n > N$ vanishes in the quotient. Evaluation is a homomorphism from the finite polynomial representative to $\mathbb{Q}$. $\square$

This is the one-generator face of $e = \exp(1)$. The value E_N is not e ; it is an exact rational construction with a known positive remainder. The analytic exponential identifies the transported limit, while the finite generator determines how the next body grows.

3.2. Prime powers are required by the Euler exponential.

Introduce one formal axis X_p for each prime p and the power sums

$$P_m = \sum_p X_p^m, \quad L = \sum_{m \geq 1} \frac{P_m}{m}.$$

Theorem 3.2.2. In the total-degree completion of the commutative rational algebra on the prime axes,

$$\exp(L) = \prod_p (1 - X_p)^{-1}.$$

Proof. The formal logarithm gives

$$-\log(1 - X_p) = \sum_{m \geq 1} \frac{X_p^m}{m}.$$

Summing over p yields L . Formal exponentiation converts the sum into the product. At every bounded total degree, only finitely many factors and powers contribute to a fixed monomial, so the identity is coefficientwise well-defined. $\square$

At degree two,

$$[\exp(L)]_2 = \frac{P_2}{2} + \frac{P_1^2}{2}.$$

The two summands have different work. For $p \neq q$, the monomial $X_p X_q$ occurs twice in P_1^2 and division by two gives coefficient one. For a square X_p^2 , the term $\frac{P_1^2}{2}$ contributes only $\frac{1}{2}$; $\frac{P_2}{2}$ supplies the missing $\frac{1}{2}$. Thus degree two is the semiprime population only when both distinct and repeated prime incidences are retained.

Corollary 3.2.3. Withholding $\frac{P_2}{2}$ leaves an exact residual supported only on the repeated-prime diagonal:

$$[\exp(P_1)]_2 - \left[\prod_{p(1-X_p)}^{-1} \right]_2 = -\frac{1}{2} \sum_p X_p^2.$$

This is not a new classification of integers. It is a coefficient-level statement about repeated traversal of one formal prime axis. Under the specialization $X_p = p^{-s}$ with $\operatorname{Re}(s) > 1$, the formal identity becomes the absolutely convergent Euler-product identity for $\zeta(s)$ [4]. The specialization does not supply analytic continuation.

3.3. Aperture growth without replay.

Write

$$L = \sum_{k \geq 1} L_k, \quad E = \exp(L) = \sum_{n \geq 0} E_n,$$

with L_k, E_k homogeneous of degree k .

Theorem 3.3.4. The next homogeneous face obeys

$$nE_n = \sum_{k=1}^n kL_k E_{n-k}.$$

Proof. Introduce a scalar t and write $L(t) = \sum_{k \geq 1} t^k L_k$ and $E(t) = \exp(L(t))$. Formal differentiation gives $E'(t) = L'(t)E(t)$. The coefficient of t^{n-1} on the left is nE_n ; on the right it is $\sum_{k=1}^n kL_k E_{n-k}$. $\square$

For the Euler atlas, $L_k = \frac{P_k}{k}$. At degree three,

$$E_3 = L_3 + L_1 L_2 + \frac{L_1^3}{6}.$$

Supplying $L_3 = \frac{P_3}{3}$ makes every degree-three monomial coefficient one. If only $\frac{P_3}{3}$ is withheld, every mixed monomial still has coefficient one while every cube X_p^3 has coefficient $\frac{2}{3}$. The complete oriented residual against the direct Euler product is

$$-\frac{1}{3} \sum_p X_p^3.$$

The pattern exposes the role of the prime-power logarithmic current at exactly the repeated-axis faces; it does not require a global recomputation of lower degrees.

For the one-generator exponential,

$$E_{N+1}(y) = E_{N(y)} + \frac{y^{N+1}}{(N+1)!}.$$

The measured aperture $8 \rightarrow 9$ used

$$E_8(1) = \frac{109601}{40320}, \quad E_9(1) = \frac{98641}{36288}.$$

The returned degree-eight body was the actual predecessor; the degree-nine term was added once, and the lower restriction remained exact.

SOURCE ROUTE. Derivation and provenance: The infinite formula is the finite generator and The aperture grows by its next homogeneous face . Exact machine evidence: Formula ecology 01 and Formula aperture growth 01.

3.4. Two exact atlases for pi.

The native geometric quantity is a complete oriented turn Θ ; the conventional half-turn constant is $\pi = \frac{\Theta}{2}$. A positively oriented polygon carries a finite sum of exterior turns. Under the usual regularity assumptions, refinement toward a winding-one curve gives the total curvature theorem

$$\sum_j \Delta\theta_j \rightarrow \int_{\gamma} \kappa \, ds = \Theta.$$

This is a geometric atlas for π . A separate arithmetic atlas uses exact rational arctangent partials.

Proposition 3.4.5. Machin's identity is

$$\frac{\pi}{4} = 4 \arctan\left(\frac{1}{5}\right) - \arctan\left(\frac{1}{239}\right).$$

Proof. Put $a = \arctan\left(\frac{1}{5}\right)$ and $b = \arctan\left(\frac{1}{239}\right)$. The double-angle formula gives $\tan(2a) = \frac{5}{12}$ and $\tan(4a) = \frac{120}{119}$. Hence

$$\tan(4a - b) = \frac{\frac{120}{119} - \frac{1}{239}}{1 + \frac{120}{119 \cdot 239}} = 1.$$

Since $0 < 4a - b < \frac{\pi}{2}$, it equals $\frac{\pi}{4}$. □

For $0 < x \leq 1$, the alternating expansion

$$\arctan(x) = \sum_{n \geq 0} (-1)^n \frac{x^{2n+1}}{2n+1}$$

has a remainder bounded in magnitude by the first omitted term, with its sign. Substituting $x = \frac{1}{5}$ and $x = \frac{1}{239}$ therefore gives exact rational upper and lower enclosures for π at every aperture. No decimal or floating geometry is needed for the causal comparison.

The Euler/Dirichlet atlas instead supplies exact finite faces of

$$\zeta(2) = \sum_{n \geq 1} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

The last equality is a classical theorem, not an identity inferred from overlap between finite enclosures. Primes are structural in the Euler path; prime factors inside Machin denominators are features of that inscription, not evidence that both paths have the same construction.

3.5. Scale and turn meet on the logarithmic cover.

For a nonzero complex path $z(t) = r(t)e^{i\theta(t)}$,

$$\frac{dz}{z} = d(\log r) + i d\theta.$$

Its lifted integral is

$$\Xi_\gamma = \int_\gamma \frac{dz}{z} = \log\left(\frac{r_1}{r_0}\right) + i(\theta_1 - \theta_0 + n\Theta).$$

The exponential presents the endpoint ratio,

$$\frac{z_1}{z_0} = \exp(\Xi_\gamma),$$

but forgets the integer winding because $\exp(w + in\Theta) = \exp(w)$. Thus e is the unit ratio of normalized logarithmic scale transport, while π is half a native turn. Euler's relation

$$\exp(i\pi) = -1$$

folds a half-turn lift to the additive inverse; it does not identify magnitude and phase as one undifferentiated quantity.

This exact scale-turn geometry will later support annular and fractal comparisons. It is not a claim that e intrinsically means compression or that π intrinsically means coherence.

3.6. Finite theta involution.

For each lattice shell $n \geq 0$, retain two formal Gaussian arms

$$d_{n(t)} = \exp(-\pi n^2 t), \quad r_{n(t)} = t^{-\frac{1}{2}} \exp\left(-\pi \frac{n^2}{t}\right).$$

Define the typed involution $Jd_n = r_n$, $Jr_n = d_n$. At aperture N ,

$$\Theta_N^d = d_0 + 2 \sum_{n=1}^N d_n, \quad \Theta_N^r = J\Theta_N^d.$$

A finite raw truncation is not self-dual. Its complete decomposition is

$$S_N = \frac{\Theta_N^d + \Theta_N^r}{2}, \quad A_N = \frac{\Theta_N^d - \Theta_N^r}{2},$$

with $JS_N = S_N$ and $JA_N = -A_N$.

Lemma 3.6.6. The aperture update

$$S_{N+1} = S_N + d_{N+1} + r_{N+1}, \quad A_{N+1} = A_N + d_{N+1} - r_{N+1}$$

preserves the invariant and anti-invariant eigenspaces exactly.

Proof. Apply J , use $Jd_{N+1} = r_{N+1}$ and $Jr_{N+1} = d_{N+1}$, and collect the two signs. $\square$

For $n > 0$, formal Mellin transport gives

$$\mathcal{M}[d_n](s) = \Gamma\left(\frac{s}{2}\right)(\pi n^2)^{-\frac{s}{2}},$$

$$\mathcal{M}[r_n](s) = \Gamma\left(\frac{1-s}{2}\right)(\pi n^2)^{-\frac{1-s}{2}}.$$

Substitution $s \rightarrow 1-s$ exchanges the receipts and fixes $s = \frac{1}{2}$. The isolated $n = 0$ endpoint requires separate analytic treatment. A finite symmetric projection proves only invariance under the declared J ; it does not prove raw theta reciprocity.

3.7. The residual is a moving boundary.

Let the complete direct and reciprocal charts be split at aperture N :

$$C_c = P_{c,N} + T_{c,N}, \quad c \in \{d, r\}.$$

If an inherited Poisson theorem identifies $C_d = C_r$, then

$$P_{d,N} - P_{r,N} = T_{r,N} - T_{d,N}.$$

The visible finite mismatch is exactly the complementary tail mismatch.

Proposition 3.7.7. Moving shell $N+1$ from each tail to its partial preserves the balance identity above. In the ordered basis $[d_{N+1}, r_{N+1}]$,

$$\Delta(P_d - P_r) = [2, -2] = \Delta(T_r - T_d).$$

Proof. Substitute $P_{d,N+1} = P_{d,N} + 2d_{N+1}$ and $T_{d,N+1} = T_{d,N} - 2d_{N+1}$, together with their reciprocal counterparts. Both sides of the balance identity receive the same coefficient vector. $\square$

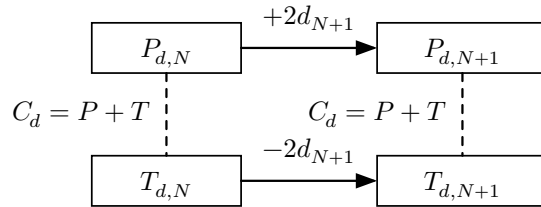


FIGURE 4. Aperture growth transfers one boundary constituent from the generative remainder into the participating face. The complete chart is unchanged.

The tail is not an infinite active population. Its finite generative description retains the next shell, multiplicity, hand, positive domain, and recurrence

$$d_{n+1} = d_n \exp(-\pi t(2n+1)), \quad r_{n+1} = r_n \exp\left(-\frac{\pi(2n+1)}{t}\right).$$

At aperture $4 \rightarrow 5$, the exponent gap carried exactly $6^2 - 5^2 = 11$. The Poisson law was inherited from classical analysis, not discovered by the machine.

SOURCE ROUTE. The finite involution and boundary-transfer measurements are Theta/Mellin aperture 01 and Poisson tail transport 01. Classical theta and zeta identities are recorded in [5] and [4].

3.8. The exact RH boundary.

The completed zeta function satisfies a functional equation under $s \rightarrow 1-s$. After centering $z = s - \frac{1}{2}$, the involution is $J(z) = -\bar{z}$ and its fixed locus is $\text{Re}(z) = 0$, equivalently $\text{Re}(s) = \frac{1}{2}$. Functional population symmetry means zeros occur in the corresponding orbits. The Riemann hypothesis is the stronger pointwise assertion that every nontrivial zero lies on the fixed locus.

The derivations above establish why the half-density appears:

$$\text{quadratic metric} \rightarrow \text{diffusion length } \sqrt{t} \rightarrow \text{reciprocal factor } t^{-\frac{1}{2}},$$

$$\text{reciprocal factor } t^{-\frac{1}{2}} \rightarrow \text{theta duality} \rightarrow \text{Mellin reflection} \rightarrow \text{fixed coordinate } \frac{1}{2}.$$

They do not force each zero onto that coordinate. Opposed off-line residuals can preserve aggregate symmetry while neither point is fixed.

Open problem 3.8.8. Construct a complete arithmetic carrier whose trace is the completed explicit formula on an admissible test space and whose relevant quadratic form is positive exactly when every nontrivial zero lies on $\text{Re}(s) = \frac{1}{2}$. Show that the finite prime-power, archimedean, aperture, and tail transitions converge to this carrier without assuming the desired positivity.

This is the honest remaining bridge toward Weil's positivity criterion [6]. A finite positive coefficient pairing, a symmetric theta projector, a finite zero census, or a recurring visual shell cannot replace it. The project therefore has a precise RH-facing programme, but no RH proof.

EVIDENCE RECEIPT. *Claim.* Finite exponential, Euler, aperture, involution, and boundary-transfer relations form one exact generator network.

Grade. Formal derivation plus bounded exact rational/integer measurement. Classical analytic laws are externally inherited.

Support. Formula §§CIII, CXLVIII-CXLIX; four exact formula-ecology results; DLMF theta/zeta identities.

Open boundary. Analytic continuation, the completed explicit-formula trace carrier, support-covariant positivity, and RH remain open.

4. J5. COMPARATIVE GEOMETRY, PHYSICS, LIFE, AND CULTURE

CONTENTS OF THIS PART

A visual grammar for relational geometry	Hypergraph rewrite and oriented comparison
Rebase, retriangulation, and physical relocation	Invariants retain the constrained directions
Annular scale-turn recurrence and fractals	Metric, curvature, and Ricci flow
Physics bridges require domain law	Lightning and living terrain
Comparative burden for a theory of everything	

The foundational calculus is deliberately thinner than any of its applications. This is a strength only if comparison remains exact about what each domain adds. We use three relation grades:

- an **exact formal match** identifies the same mathematical expression under a stated map;
- a **direct correspondence** preserves a named structure while leaving other structures outside the map; and
- a **structural resonance** is a hypothesis-generating analogy with a stated non-equivalence.

The grades do not form an automatic ladder. A beautiful structural resonance does not become an exact match through repetition.

4.1. A visual grammar for relational geometry.

The journal uses four visible primitives. A node is an actual occurrence only when the source supplies that occurrence. A line is admitted incidence or transport only when its direction and role are declared. A polygonal face records boundary, convex participation, or comparison according to its caption; it never acquires all three readings automatically. A dashed line marks a quotient, proposed bridge, world passage, or OPEN comparison and must be named locally.

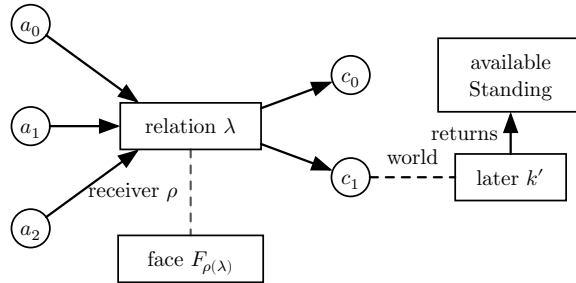


FIGURE 5. Relational diagram grammar. Antecedents meet in one actual relation occurrence; immediate consequents emanate; only a later world return can re-enter as Current. The dashed receiver edge is a presentation of the relation, not another causal owner.

Figure 5 is the general antecedent–consequent grammar. The relation occurrence, rather than an arrow floating between private objects, is the middle of the diagram. This accommodates arbitrary-arity hyperedges while retaining the distinction between event, causal dependency, and receiver presentation [7].

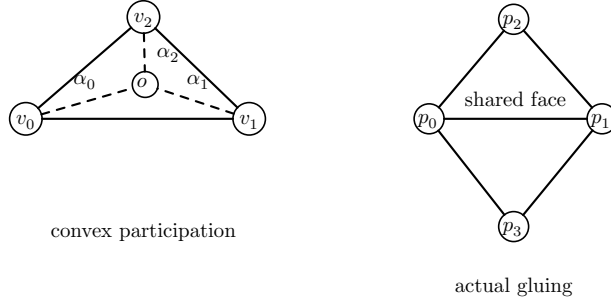


FIGURE 6. Two structures which drawings often blur. Left: a receiver output is a barycentric point of the admitted values when the coefficients are nonnegative and sum to one. Right: a simplicial complex requires actual common-face incidence; geometric overlap alone is insufficient.

Figure 6 separates a convex simplex from a simplicial complex. The former is a geometric realization with barycentric points; the latter additionally requires closure under faces and actual common-face intersections [8–10]. This distinction is central to the attention comparison in J3: normalized positive coefficients give a convex point, but affine dependence can make the realized face degenerate.

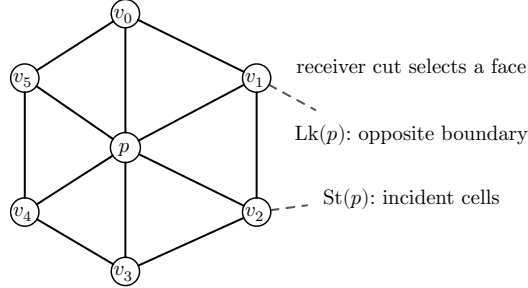


FIGURE 7. A local star and link. The star records the simplices incident to the selected occurrence; the link records the boundary seen opposite it. Either is a useful receiver-local chart, but neither is the complete ecology.

Figure 7 adds the situated camera. A star answers which cells meet the selected occurrence; a link answers what lies opposite that occurrence in those cells. They are useful local views of a complex, not observer-independent identities of the whole. This is the visual discipline behind sheet, seam, pin, fold, and receiver cross-section in the recent derivations.

Remark. Figures in this volume organize exact relations already stated in symbols or source data. Their coordinates, apparent distances, and visual symmetry are expository geometry and cannot become causal evidence.

4.2. Hypergraph rewrite and oriented comparison.

A conventional cellular automaton assigns local state to sites on a fixed lattice and evolves by a chosen neighborhood rule. Triangular cells do not change that

ontology by themselves. The candidate Eros object is instead occurrence-relational: arbitrary-arity relations determine local contact; actual update events replace relations; oriented incidence determines boundary; causal dependency records which event used which material; and transport compares paths.

The Wolfram model makes a useful external distinction among spatial hypergraph, updating event, causal graph, and multiway graph [11]. The first three have direct structural correspondences here. A multiway system is an analysis of possible histories, not a requirement that every possible match remain active inside Soma. Causal invariance is an additional confluence obligation and cannot be inferred from deterministic scheduling.

Notation	Meaning and jurisdiction
spatial hyper-graph	one relation cut; arbitrary local arity
rewrite event	actual local replacement under an admitted rule
causal graph	dependency among actual events
multiway graph	analytical population of alternative update histories
oriented complex	dimension, orientation, shared boundary, and $\partial^2 = 0$
transport	what the admitted path does to a carrier

TABLE 2. Core notation. Every symbol is indexed by a declared receiver, frame, or boundary when that index can change the result.

The triangle is the smallest cell where boundary closure and route comparison can coexist:

$$\partial\tau = a_{12} - a_{02} + a_{01}, \quad X_\tau = (T_{a_{12}} \circ T_{a_{01}}, T_{a_{02}}).$$

But the two readings remain distinct. Boundary closure follows from oriented incidence; path commutation follows from the transport. A shared edge can also serve as a preserved interface for a rewrite. None of these roles follows from three lines merely intersecting in a drawing.

SOURCE ROUTE. The full typed comparison is The hypergraph supplies the rewrite; the triangle carries the comparison. It explicitly corrects the earlier tendency to use a strict category nerve or a triangular cellular automaton as the universal core.

4.3. Rebase, retriangulation, and physical relocation.

Three operations have often appeared under the single phrase “change of basis”:

1. **carrier rebase:** local coordinates change by invertible maps G_x , while incidence and the exterior relation remain fixed;
2. **retriangulation:** internal cells or factorization change inside one preserved exterior boundary; and
3. **physical relocation:** residency, communication geometry, length, or energetic cost changes.

In two dimensions, a $2 \rightarrow 2$ Pachner move exchanges the diagonal of a quadrilateral while holding its exterior boundary. The $1 \rightarrow 3$ and $3 \rightarrow 1$ moves introduce or remove one interior vertex. These are exact PL-topological operations under Pachner's hypotheses [3]. They do not preserve a chosen metric automatically, and they do not decide which later consequence matters.

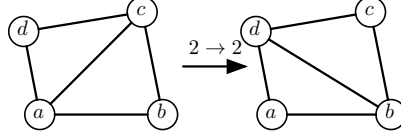


FIGURE 8. The two-dimensional bistellar move. The exterior quadrilateral is retained while the interior diagonal changes. Preserving topology is supplied by the move's hypotheses; preserving transport, metric, or physical cost requires separate comparison.

Proposition 4.3.1. If a carrier rebase assigns an invertible map G_x at every object and replaces edge transport $T_a : x \rightarrow y$ by

$$T'_a = G_y T_a G_x^{-1},$$

then every path comparison transforms covariantly:

$$T'_\gamma = G_y T_\gamma G_x^{-1}.$$

Hence equality of parallel transports is preserved.

Proof. Along a composable path the adjacent factors $G_z^{-1} G_z$ cancel. Both parallel paths acquire the same left factor at the endpoint and right factor at the source, so one pair is equal exactly when the rebased pair is equal. $\square$

This proposition is the rigorous gauge face. A retriangulation needs a different proof: the internal factorization must be replaced and its exterior transport shown to commute. Physical relocation needs measurements of communication, memory movement, latency, or energy. Treating all three as algebraic gauge hides the most interesting empirical question.

4.4. Invariants retain the constrained directions.

Let (M^N, g) be a Riemannian manifold and $I : M \rightarrow \mathbb{R}^k$ a regular constraint map. The active tangent space of the level set is

$$D_x = \left\{ v : dI_x^{a(v)} = 0 \text{ for every } a \right\}, \quad \dim D_x = N - k.$$

The constrained directions have not vanished. They remain as normal directions, the value $I(x)$, a coarea Jacobian, an orientation, and the receipt needed to sweep the fibers.

For the Euclidean N -ball,

$$V_{N(R)} = c_N R^N, \quad c_N = \frac{\pi^{\frac{N}{2}}}{\Gamma(\frac{N}{2} + 1)},$$

and its boundary hyperarea is

$$A_{N-1}(R) = Nc_N R^{N-1} = \frac{dV_N}{dR}.$$

Consequently,

$$V_{N(R)} = \int_0^R A_{N-1}(r) dr.$$

The final sphere does not reconstruct a labeled ball interior; the ordered radius family and the normal sweep do. The Gamma recurrence gives

$$V_{N+2}(R) = 2\pi \frac{R^2}{N+2} V_{N(R)}, \quad A_{N+1}(R) = 2\pi R V_{N(R)}.$$

The visual intuition developed in 3Blue1Brown's high-dimensional sphere lesson is therefore coupled here to an exact radial FTC and Gamma calculation, not used as proof authority [12].

Remark. The exponent $(\lambda^N)^{\frac{N-1}{N}} = \lambda^{N-1}$ extracts codimension-one scale from an isotropic cube. For a ball, the coefficient c_N must be retained. Equal scaling exponents do not identify cube and sphere constructions.

The coarea theorem supplies the general form:

$$\int_M h(x) J_{I(x)} dV_N = \int_{\mathbb{R}^k} \left(\int_{I^{-1}(c)} h dA_{N-k} \right) dc.$$

This is an exact mathematical model of “a family of cuts carries a volume.” It becomes a holonic application only when the receiver, constraint, orientation, and event which performs the hand-up are also supplied.

4.5. Annular scale-turn recurrence and fractals.

The logarithmic lift from J4 has the geometry

$$w = u + i\theta, \quad z = \exp(w) = e^u e^{i\theta}.$$

An annulus is the quotient

$$A(r, R) \equiv \frac{\{\log r < \operatorname{Re}(w) < \log R\}}{i\Theta\mathbb{Z}},$$

with conformal modulus

$$\operatorname{mod} A(r, R) = \frac{\log\left(\frac{R}{r}\right)}{\Theta}.$$

It retains a radial scale interval and a cyclic return. Exponentiation presents one nonzero face while identifying lifts separated by a complete turn.

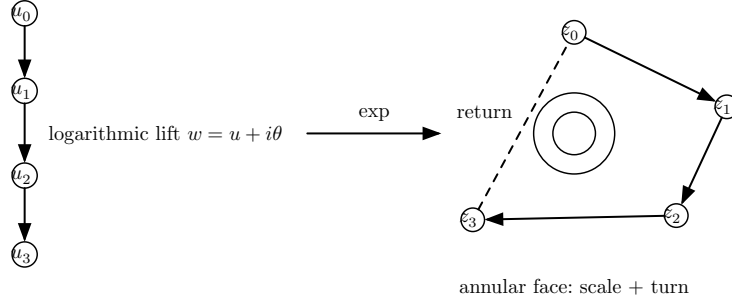


FIGURE 9. The scale-turn chart. Translation in the logarithmic lift becomes simultaneous radial scaling and angular return under exponentiation. A drawn recurrence becomes evidence only when the return map, receiver landmarks, multiplier, and residual are supplied.

Definition 4.5.2. A **self-similar return at a declared cut** consists of a first-return map F^q , receiver landmarks, a chart ψ , a multiplier λ , incidence or winding, and the residual of the actual return. Visual resemblance alone is not a return law.

Near a suitable noncritical periodic point, Koenigs linearization may give

$$\psi(F^{q(z)}) = \lambda\psi(z), \quad \lambda = \exp(\kappa + i\omega).$$

Then

$$\psi(F^{qN}(z)) = e^{N\kappa} e^{iN\omega} \psi(z).$$

Here κ is logarithmic scale change and ω is turn per return. In higher dimensions the transport is generally ordered,

$$v_N = J_N \dots J_2 J_1 v_0,$$

and becomes a scalar power only in a commuting one-dimensional chart. Tan Lei's local asymptotic similarity between Mandelbrot and Julia sets supplies a rigorous example of receiver-local recurrence [13]; it does not license generic exact pixel copies.

| SOURCE ROUTE. The scale-turn derivation, its corrections, and the fractal boundary are in The annulus carries the scale-turn lift .

4.6. Metric, curvature, and Ricci flow.

For a declared Riemannian receiver (M, g) , the metric compares tangent directions; the Levi-Civita connection transports them; and the Riemann tensor measures noncommuting infinitesimal transport. For a geodesic tangent u and separation ξ ,

$$\text{Gravitas}_{\gamma(\xi)} = \nabla_u \nabla_u \xi = -R(\xi, u)u$$

under the project's chosen sign convention. With an orthonormal transverse frame $\{e_a\}$,

$$\sum_a g(\text{Gravitas}_{\gamma(e_a)}, e_a) = -\text{Ric}(u, u).$$

Ricci is the transverse trace of directional curvature, not the whole curvature operator.

Hamilton's Ricci flow

$$\partial_\tau g = -2 \operatorname{Ric}(g)$$

changes the comparison law itself [14]. This is an exact physical-mathematical specialization after a Riemannian metric and its PDE have been supplied. It is not a universal Soma update law. Perelman's Ricci-flow programme supplies a profound local-to-global example: smooth evolution, curvature concentration, blow-up and scale rebase, controlled neck surgery, component departure, continued flow, and topological accounting establish the Poincare result [15, 16].

The distinctions matter:

- smooth Ricci flow preserves topology while it exists;
- surgery is the typed topological deed;
- normalization is a scale rebase;
- a soliton is return up to diffeomorphism and scaling; and
- extinction is not the same receiver as normalized persistence.

Poincare does not imply $P = NP$, universal confluence, or trivial connection holonomy. It is a rigorous example in which local evolution and controlled singular transformation expose a global invariant without enumerating all presentations.

4.7. Physics bridges require domain law.

In a nonlinear sigma model, the target metric is a coupling and the leading metric beta function contains the Ricci tensor, subject to scale conventions, field redefinitions, and higher corrections [17]. Under its Abelian-isometry hypotheses, T-duality exchanges circle momentum and winding and acts on the full background, not only the radius. Flow, singularity, rebase, duality, and return are therefore different deeds.

For an unwarped product $M_{11} = M_{10} \times S_R^1$,

$$dV_{11} = R d\theta dV_{10}, \quad V_{11} = 2\pi R V_{10}.$$

Dimensional reduction is fiber integration. At low enough receiver energy the translation-invariant zero mode may be active while nonzero modes retain momenta $\frac{n}{R}$. The circle direction is dormant at that receiver, not deleted from the construction. Witten's Type-IIA/eleven-dimensional relation supplies a typed physical instance of this principle [18].

No such correspondence makes software arcs physical strings or makes receiver-relative compression a law of spacetime. A physical bridge owes at least:

1. mathematical objects and maps;
2. dimensions and units;
3. dynamical law and admissible initial/boundary conditions;
4. limiting agreement with established theory; and
5. a discriminating observation not already inserted as source structure.

4.8. Lightning and living terrain.

Lightning is a materially grounded example of a path changing the terrain later current receives. A leader changes ionization, space charge, conductivity, temper-

ature, and channel geometry; later leaders and the return stroke propagate through that changed medium. The structural sequence is

field + atmosphere $\rightarrow$ leader founding $\rightarrow$ changed conductive terrain,
 changed terrain $\rightarrow$ later ride or new founding $\rightarrow$ attachment,
 attachment $\rightarrow$ finite return $\rightarrow$ material consequence.

This is a direct correspondence to the event/Standing/consequence circuit at the causal grain. The literal law remains Maxwell, plasma, chemical, thermal, and gas-dynamic physics. Channel curvature, an effective transport metric, Ricci curvature, and spacetime curvature are four different receiver faces.

The same discipline applies to biology and culture. Niche construction, development, ecological inheritance, distributed cognition, ritual, language, and institutions can preserve relations through changing constituents. They become holonic comparisons only after identities, transmission channels, receiver cuts, causal contrasts, and failure cases are specified. Narrative continuity is not automatically a biological lineage; cultural resemblance is not automatically common ancestry; and a useful metaphor is not a mechanism.

SOURCE ROUTE. The geometry/physics chain is developed in The hyperarea is the invariant fiber and The Ricci trace changes the receiver . Their strongest equations are retained here; their broader “one chain” language is narrowed to typed correspondences.

4.9. Comparative burden for a theory of everything.

The hypothesis of everything is best stated as a research posture: one relational grammar may coordinate mathematics, computation, life, cognition, and physics without making any application the definition of the grammar. At present, the programme has:

- a receiver-relative event calculus;
- exact local constructions in several mathematical charts;
- a production machine with bounded measurements; and
- disciplined bridges to established domain theories.

It does not yet have a physical action, unit system, quantum/statistical limit, experimentally novel prediction, or falsification result spanning those domains. “Relational unification programme” is therefore the strongest licensed description. The missing physical relations are scientific work to perform, not rhetorical gaps to close.

EVIDENCE RECEIPT. *Claim.* The same event calculus admits several rigorous specializations without identifying them.

Grade. Exact matches and direct correspondences where maps are written; structural resonance elsewhere.

Support. Hypergraph/rewrite, coarea, annulus, Ricci-flow, compactification, and lightning source routes.

Open boundary. No universal manifold, field equation, biological mechanism, cultural taxonomy, or physical theory of everything is established.

5. J3. LEARNING, INHERITED ECOLOGY, AND MODERN ML

CONTENTS OF THIS PART

Persistence classes are not one memory	A network is one factorization
Exact inherited law after source departure	Contextual factorization and its boundary
Cultivation: changed later conduct and interference	Completion into a local transformation
Partial translation atlas	Parent-on-OPEN substitution
Probability, loss, and learning	

Modern machine learning supplies lawful inherited technology: tokenizers, embeddings, trained weights, attention, parsers, optimizers, and datasets may enter as Current or Standing when their source and role are declared. The scientific task is not to pretend they are absent. It is to distinguish what was inherited, what the event formed, what later stood, and what later conduct changed.

5.1. Persistence classes are not one memory.

A tokenizer is a boundary chart

$$T_\tau : C \rightarrow V^*$$

which segments an ordered carrier and emits codebook addresses. A token is neither a word nor a semantic atom. The chart chooses grain, chronology length, adjacency, and computational cost before learned conduct begins.

For one scaled dot-product attention receiver,

$$\kappa_{i,j} = \exp\left(\frac{q_i \cdot k_j}{\sqrt{d}}\right), \quad \alpha_{i,j} = \frac{\kappa_{i,j}}{\sum_m \kappa_{i,m}},$$

$$o_i = \sum_j \alpha_{i,j} v_j.$$

Query and key expose contact; value carries what crosses; softmax normalizes the admitted local population. The weights are deterministic receiver quotients, not truth probabilities. Causal masking is an authored chronology boundary. A key/value cache is episode state, not trained Standing or complete lineage.

Proposition 5.1.1. If the values $v_1, \dots, v_n$ lie in a real affine space, then the softmax output o_i lies in their convex hull. It has unique barycentric coordinates exactly when the admitted values are affinely independent.

Proof. Every $\alpha_{i,j} > 0$ and $\sum_j \alpha_{i,j} = 1$, so o_i is a convex combination. Barycentric coordinates relative to the listed vertices are unique exactly when no nontrivial affine dependence preserves both the coefficient sum and the resulting point. $\square$

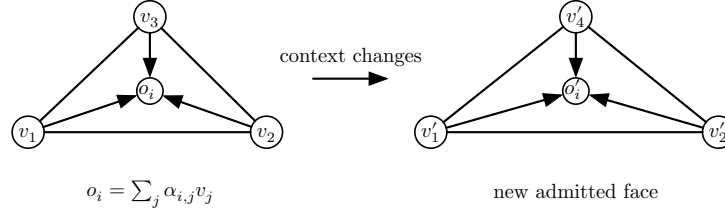


FIGURE 10. Attention as event-local barycentric transport. Context can change the query, keys, values, admitted population, and therefore the realized simplex itself. The picture does not identify a token, value vector, neuron, or attention coefficient with a stable meaning.

The attention face is therefore a genuine simplex only under affine independence. Context can change the vertices, the coordinates, and the candidate population. A zero or negligible coefficient restricts one event-local face; it does not delete a universal constituent. The staged key–query, relevance-grid, weighted-value, and residual presentation used by 3Blue1Brown is especially effective because each visual answers one operation before the view zooms back out to the whole block [19]. Figure 10 adopts that explanatory discipline while keeping the journal’s stricter occurrence and receiver distinctions.

Residual depth recurrence

$$h_{\ell+1} = h_{\ell} + \Delta_{\ell(h_{\ell})}$$

carries a changing activation state. Parameters, activations, optimizer state, recurrent state, episode cache, and external world material are different persistence classes. Calling all of them “memory” erases the causal cut which a learning claim needs.

SOURCE ROUTE. This comparative ontology is fixed in Formula §CXXVI and its complete source map. The token is a boundary address . The transformer equations follow [20]; their holonic interpretation is a project derivation, not a claim made by that paper.

5.2. A network is one factorization.

A trained configuration realizes a contextual relation

$$F_{\theta} = f_L \circ \dots \circ f_1.$$

Its channels, layers, residual branches, and expert boundaries belong to that realization. Even a linear factorization has gauge:

$$BA = (BG^{-1})(GA)$$

for invertible G . Nonlinearities restrict lawful transformations, but permutations, compensated scalings, and function-preserving architecture transformations still separate realized function from internal coordinate identity [21].

Definition 5.2.2. A **neural retriangulation** is a change of internal factorization which preserves a declared exterior relation on a named receiver family. It is stronger than coordinate rebase and independent of physical redistribution.

To establish such a retriangulation, equality on a training sample is insufficient. The two factorizations need a declared overlap or test family, complete exterior

comparisons, structured residuals, and physical measurements appropriate to the claim. If latency or energy is part of the thesis, algebraic equivalence alone cannot supply it.

The fixed-basis OLMo observations below should therefore be read as exact coordinate censuses, not discoveries of invariant experts or detachable skills.

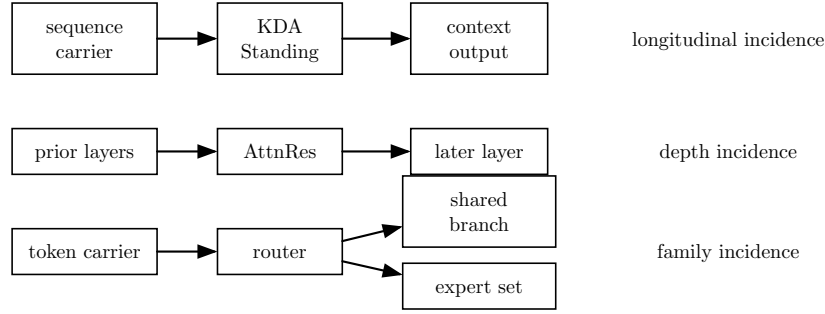


FIGURE 11. Three independent architectural incidence axes in the public Kimi K3 account: longitudinal state across sequence, selective residual transport across depth, and sparse transformation-family routing. The diagram is a typed comparison, not a claim that these mechanisms are one universal holonic primitive.

Figure 11 prevents the word “attention” from collapsing three independent directions. The public Kimi K3 account distinguishes longitudinal Kimi Delta Attention, selective residual transport across depth, and sparse Stable LatentMoE routing across transformation families [22]. That modern architectural overview is a useful visual analogue for holonics because it shows several incidence axes at once; it is not evidence that the architecture realizes Eros. Likewise, distributed feature directions and superposition must not be redrawn as one concept per neuron [23].

5.3. Exact inherited law after source departure.

The first law-ecology experiment carried two exact dyadic linear laws into cellular Standing: a calibration matrix and a 2×2 restriction from OLMo 2. Source lineages departed. Held-out currents later recruited each law from public radiation and produced the exact product. No-Standing, wrong-interface, and retired-interface foils refused. A declared oriented rebase replaced each law; the successor conducted after exact rest/remount.

For the calibration:

$$M = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \quad x = \begin{pmatrix} 5 \\ 8 \end{pmatrix}, \quad Mx = \begin{pmatrix} 13 \\ 21 \end{pmatrix}.$$

After the declared right rebase,

$$M' = \begin{pmatrix} 1 & 2 \\ 1 & 3 \end{pmatrix}, \quad M'x = \begin{pmatrix} 21 \\ 29 \end{pmatrix}.$$

This closes source-to-Standing carriage and later source-absent conduct at the cut. The update law was supplied; the machine did not infer a general rebase.

5.4. Contextual factorization and its boundary.

For OLMo 2 layer 7, one fixed-basis factor contribution is

$$g_{j(x_c)} = \text{SiLU}\left((W_{\text{gate}}x_c)_j\right)(W_{\text{up}}x_c)_j,$$

$$E_{c,j,a} = g_{j(x_c)}W_{\text{down}[a,j]}, \quad y_{c,a} = \sum_{j=0}^{8191} E_{c,j,a}.$$

An exact census on five receiver axes selected six coordinate summands across three contexts. Four were common; prose exchanged one summand. The same five-summand membership in conversation and code-comment still produced different exact outputs because contemporary activations differed.

The important receiver dependence is numerical:

Context	sign	order	sign + abs.	BF16
conversation	5	1,936	5,092	7,819
prose	5	2,775	5,557	7,652
code comment	11	5,017	5,017	7,858

TABLE 3. Smallest stable population size for four nested receiver faces in one fixed OLMo factorization. Exact dyadic identity required all 8,192 summands in every context.

The five-summand aperture preserved only the orientation $(-, -, -, +, +)$. It did not preserve magnitudes or axis order. The approach to the stricter boundaries was nonmonotone. This is evidence against identifying one small coordinate cut with a context-independent skill.

SOURCE ROUTE. Exact evidence: Transformer contextual hexis 01. Its original “expert” interpretation is explicitly superseded by The network is one realization; the relation may be retriangulated . The arithmetic remains accepted.

5.5. Cultivation: changed later conduct and interference.

An online cultivation passage began with the six fixed-basis factors and admitted at most 32 new factors when a declared five-axis orientation-plus-order face remained OPEN. Over nine currents, 15 new factors entered Standing, producing 21 of 8,192 factors. Two later currents closed with no source return even though the original six-factor body remained OPEN:

Later current	original six	then-Standing	new retrieval
prose - relevant after	OPEN	closed under 19	0
code comment - ir-relevant before	OPEN	closed under 20	0

TABLE 4. Matched learning contrast. Earlier returned factors participated in later closure.

This satisfies the J1 learning criterion at the declared face. It also revealed interference: two currents which would close under the original six were reopened

by intervening cultivation and needed one further factor. Growth was contextual deformation, not monotone accumulation of universally helpful parts.

Three frozen changed-operation holdouts preserved only the coarse orientation; signed order remained OPEN. Selection scanned 122,685 candidates and upstream OLMo still supplied the full activation field. The result is cultivation evidence, not a portable skill or efficiency result.

5.6. Completion into a local transformation.

A later experiment formed a deterministic rank-six chart. Let $G \in \mathbb{Q}^{6 \times 8192}$ be six exact activation rows, choose the first rank-raising pivot columns $P = [0, 1, 2, 3, 4, 5]$, put $A = G[:, P]$, and let $Y \in \mathbb{Q}^{6 \times 5}$ be the receiver output. Exact elimination formed the unique local law

$$AK = Y, \quad K \in \mathbb{Q}^{6 \times 5}.$$

The six lower training rows crossed, were recovered from Standing, and then handed into one higher constituent carrying K . Their data and recruitment interfaces departed.

The law reproduced all six training rows, as square interpolation guarantees. The informative contrast was later context:

Population	currents	RIDE	OPEN
training conversation/prose	6	6	0
related code-comment	3	1	2
changed operation order	3	0	3

TABLE 5. Exact receiver comparison for the completed rank-six local law.

Only one related holdout RIDEd. Every other holdout retained a complete exact rational residual, with zero fabricated FOUND. The experiment therefore demonstrates local completion and later participation, not global architectural equivalence.

5.7. Partial translation atlas.

Conversation and prose also supplied 74 partial maps across 102 intervention sites where they agreed exactly at either an emitted-lineage receiver or a declared-world-affordance receiver. The code-comment chart was held out during map formation. After source departure, the later query recovered all 74 maps and invented none at 28 unsupported sites. At the held-out boundary:

Receiver	maps	RIDE	OPEN
emitted lineage	38	34	4
declared-world affordance	68	64	4

TABLE 6. Held-out translation-atlas result. The union contains 74 sites, 66 commuting maps, and 8 maps with at least one exact OPEN bit.

The atlas is local in the strict sense: repeated consequence across two charts became Standing; a third chart either conducted through it or exposed its residual. Site identity and consequence faces were source-supplied. Cross-model alignment and grounded downstream conduct remain open.

5.8. Parent-on-OPEN substitution.

The rank-six relation was then placed before the full retained parent projection

$$P_{R(g)} = \sum_{j=0}^{8191} g_j W_{R,j}.$$

The local law could substitute only when the deepest applicable Standing receiver was singleton and its face matched the derived face. Otherwise the complete parent was called once and returned the exact residual

$$r = P_{R(g)} - E_{R(g_P)}.$$

In a fixed six-current passage, one held-out receiver RIDed before parent evaluation and five remained OPEN. The exact work at that receiver was:

Path	products	accumulations
always-parent foil	245,760	245,760
selective parent	204,800	204,800
derived local work	180	180
omitted parent work	40,960	40,960

TABLE 7. Exact rational work at the declared five-axis parent receiver.

After the five residual returns, six independent source-absent probes recovered the receiver faces without another parent call. This is an operational use of OPEN: it returned to the source relation and cultivated later support. It is not end-to-end model acceleration. Upstream activations, the full MLP output, decoding, CUDA, latency, memory traffic, and energy were outside the measured cut.

SOURCE ROUTE. The evidence chain is Self-emanated law 01, Pretrained ecology cultivation 01, Contextual retriangulation 01, Transformer translation atlas 01, and Parent-on-OPEN substitution 01.

5.9. Probability, loss, and learning.

For a declared local continuation population, a recurrence quotient

$$Q_{D(\tau)} = \frac{N_{D(\tau)}}{\sum_u N_{D(u)}}$$

describes supported continuation at receiver D . It is not a substance of chance or truth. A cross-entropy scalar can be a useful differentiable receiver residual, but the causal object is the mismatch between afforded and actual later continuation. Backpropagation carries the adjoint of a performed differentiable construction; an optimizer applies a declared metric and update law; sampling is another world deed.

This yields a typed learning circuit:

$$\text{grain} \rightarrow \text{address} \rightarrow \text{transport} \rightarrow \text{transformation},$$

transformation $\rightarrow$ outgoing field $\rightarrow$ later event $\rightarrow$ residual,

residual $\rightarrow$ adjoint return $\rightarrow$ parameter deformation $\rightarrow$ changed conduct.

Conventional ML fixes much of this ecology in advance and learns within it. Eros asks which of these relations can themselves form, complete, depart, and later recur. The question is comparative, not a ban on inherited machinery.

EVIDENCE RECEIPT. *Claim.* Inherited pretrained structure can become Standing, participate in later contextual conduct, and support receiver-relative fallback.

Grade. Exact dyadic/rational bounded measurements with matched controls.

Support. Six current observation records plus the fixed-basis interpretation correction.

Open boundary. No invariant skill decomposition, learned universal router, broad transfer, English response policy, model-integrated acceleration, autonomous objective, or end-to-end efficiency is established.

6. J2. MACHINE REALIZATION AND EXACT EVIDENCE

CONTENTS OF THIS PART

Production boundary	World consequence and joint rest
Parameterized elementary mechanics	OPEN comparison becomes Standing
Conditional consequence charts	Outgoing factor and physical departure
Evidence matrix	

The theory-to-structure relation is:

Current + Standing $\rightarrow$ one machine-owned transition,
 machine-owned transition $\rightarrow$ successor + immediate radiation.

The host or CUDA executor supplies physical conduct **inside** that transition. It does not receive a receipt and reconstruct the successor. A world may mutate material and later present the consequence as new Current. Journals, hashes, observers, forests, and reports testify after the commit; they are not causal owners.

6.1. Production boundary.

The production path is

```
life::direct::current_world
-> soma_membrane::LiveCurrentMachine::receive_with
-> host or CUDA executor inside the atomic call
-> machine-validated successor
-> ContemporaryRadiation
-> possible later world consequence
```

The machine owns validation, same-prestate enactment, successor preparation, commitment of Standing and lineages, departure, and radiation. The executor may discover that a physical aperture is insufficient and request a larger one. A retry is lawful only while the predecessor is unchanged and nothing has committed.

Proposition 6.1.1. Suppose an event attempt is a pure physical evaluation of fixed (S_t, K_e) until the machine validates and commits. If every failed aperture attempt leaves S_t unchanged, then any number of resource retries followed by one successful commit is observationally equivalent at the machine boundary to one sufficiently provisioned attempt.

Proof. Every attempt evaluates the same arguments. Failed attempts have no visibility edge and make no state change. The first validated attempt determines the only committed successor and radiation, so the retry history is absent from the causal machine state except as optional testimony. $\square$

This is not a general claim about arbitrary side-effecting retries. It is a consequence of the specific uncommitted-event boundary.

6.2. World consequence and joint rest.

In the native world-rest cell, a declared forward hand changed a named register. The mutation returned as genuinely later native Current. Before that current crossed, checkpointing was refused: the machine body alone was not joint rest. The returned current afforded no second action, after which the combined world and

machine could rest. Waiting at rest launched nothing; remount and later wake were exact on host and card.

Reverse contact remained OPEN and independent contact formed no relation. Neither invented a world action. The accepted card run reported ten physical launches for nine logical currents; the tenth was one uncommitted resource retry. Host and card agreed on radiation, Standing, lineage, world state, rest, remount, and wake.

EVIDENCE RECEIPT. Claim. World mutation, genuinely later return, joint rest, silence, remount, and wake form one exact bounded circuit.

Grade. Measured host/CUDA implementation fact.

Support. Eros native world rest 01.

Open boundary. The cell did not establish durable whole-world encoding, device-resident cellular Standing, plural simultaneous world actions, sustained ecology, or scale.

6.3. Parameterized elementary mechanics.

One production test carried both:

- an oriented triangular junction with $\partial^2 = 0$; and
- a topologically dimension-zero local rewrite whose dependency rank and constituent grain changed independently.

Regional seams formed only through source-declared interface capabilities. Complete parallel paths were first-class; an optional residual did not replace them. The outgoing constituents reproduced the exact later-contact boundary after compression.

Measured quantity	value
triangle cells / incidences	21 / 27
rewrite cells / incidences	12 / 6
regional cells	4
Standing constituents	2
contact launches / lanes	6 / 57
lineage launches	9
rest words	8,876

TABLE 8. Selected exact output from Parameterized elementary mechanics 01. One worker, eight workers, CUDA, and durable rest returned equal machine faces.

The result establishes one triangle world and one rewrite world at a bounded cut. It does not establish a universal matcher, arbitrary confluence, device-resident cellular bodies, a metric or diffusion law, or arbitrary future-observation congruence.

6.4. OPEN comparison becomes Standing.

The production carrier stores a noncommuting comparison as both complete paths plus its optional exact residual. OPEN is therefore material structure rather than a flag or blacklist. In one host cell, an equal-hand opening crossed native rest and a later exact matching boundary RIDed without rewriting the earlier scope. In another, earlier OPEN Standing participated in a later actual regional contact,

changed the successor relative to a no-opening sibling, and still remained OPEN because the new comparison did not commute.

The rejected preconditions are scientifically important. A second timestamp alone did not create incidence. After real later contact was supplied, mismatch was not tuned away or regraded as FOUND. This separates chronology, incidence, comparison, and closure in the running machine.

SOURCE ROUTE. Exact carrier and runtime evidence: Open comparison Standing 01. The complete membrane suite passed 94 of 94 tests at that cut; host and CUDA agreed on the bounded mechanics cell. These counts support the implementation boundary and do not substitute for a general theorem.

6.5. Conditional consequence charts.

A source-neutral chart law received ordered categorical interfaces and a later consequence taxon. At each semantic event:

source interfaces meet prior chart Standing

- > recurrence distribution returns
- > genuinely later actual taxon arrives
- > exact membership residual stands
- > one reduced replacement chart returns

The actual consequence was not smuggled into the earlier prediction. A singleton match RIDEd; plural support containing the actual remained OPEN_INCLUDED; absence of the actual remained OPEN_RESIDUAL; and no case fabricated FOUND.

After twelve events the two operation regions had the same marginal recurrence distribution $\frac{1}{3} : \frac{2}{3}$ but different conditional interiors:

```
operation
|- drop_then_reverse  1/3 : 2/3
|  '- condition
|    |- absent        -> A
|    |- irrelevant_before -> B
|    '- relevant_after  -> B
'- reverse_then_drop  1/3 : 2/3
   '- context
      |- code_comment  -> A
      |- conversation  -> B
      '- prose         -> B
```

Collapsing either region to its marginal quotient destroys the relation needed for later conduct. This is an exact example of why compression must be indexed by a receiver family rather than a scalar notion of simplicity.

6.6. Outgoing factor and physical departure.

The first chart experiment replaced the active interfaces but retained a growing interior: 191 cells, 5,100 incidences and pins, 5,111 paths, and 5,195 transport terms in the final constituent. A follow-up let the source declare the updated chart boundary as the outgoing factor of the **joint event**. The machine committed it only after:

1. all regional contacts formed;
2. exact touched Standing-before entered joint closure;
3. the selected subcomplex passed the existing boundary-compression certificate; and

4. its exposed boundary was structurally identical to the complete joint event's exposed boundary.

An unmatched, OPEN, FOUND, or oriented residual arm would remain exposed and make the fourth check refuse.

Theorem 6.6.2. Let $j : C \rightarrow E$ select a subcomplex of the complete joint event and suppose its exposed boundary equals the exposed boundary of E . Then replacing E by C preserves every later contact whose admission factors only through that exposed boundary.

Proof. Such a later contact can inspect only the shared exposed boundary. By hypothesis the boundary occurrence population, orientation, capabilities, and admitted transport family are identical. The later contact therefore receives the same input relation. The statement makes no claim about an observation which can access an omitted interior. $\square$

The implemented check is structural, not a later garbage collector or observer reconstruction. The final active body changed from

Carrier	whole composite	outgoing factor
cells	191	3
incidences / pins	5,100 / 5,100	168 / 168
paths	5,111	168
transport terms	5,195	168
Standing scalar cells	36	36
live source lineages	0	0

TABLE 9. Bounded physical departure at the final consequential-chart cut.

Across the chronology, chart path counts could also contract ($136 \rightarrow 75$ and $140 \rightarrow 75$), so physical topology followed the successor rather than elapsed event count. All checkpoint probes, conditional relations, exact rest, and source departure remained unchanged.

SOURCE ROUTE. The two-stage evidence is Residual chart cultivation 01 and Consequential successor factor 01. The first exposes the physical-interior problem; the second closes it for the current source-declared boundary family.

6.7. Evidence matrix.

Relation	strongest support	jurisdiction
atomic successor owner-ship	source + tests	production LiveCurrentMachine
host/CUDA exact contact	measurement	two bounded mechanics worlds
world return and rest	measurement	one register world and later wake
OPEN route carriage	source + measurement	complete exposed comparisons
conditional chart cultivation	measurement	twelve source combinations
physical departure	measurement	declared outgoing boundary family
general confluence	OPEN	arbitrary interacting rewrite rules
device-resident body	cellular OPEN	CUDA forms contact, host composes topology
broad learning or agency	OPEN	not measured by these machine cells

TABLE 10. Implementation evidence does not propagate beyond the receiver named in the final column.

The current capability envelope is significant but bounded: directed hand, same-prestate successor, immediate radiation, actual world mutation and later return, silence, rest/remount/wake, plural consequences, host and CUDA contact formation, cellular Standing, exact local recurrence, receiver-relative comparison, and one source-declared physical departure family. It does not establish English, response policy, autonomous objectives, broad transfer, general agency, $P = NP$, or RH.

EVIDENCE RECEIPT. Claim. The production machine realizes the foundational event boundary and several exact later-return lifecycles.

Grade. Implemented and measured at named host/CUDA cuts.

Support. Direct-owner research deposit and five decisive observation records routed through the current README.

Open boundary. General confluence, global scale, all-card topology, device-resident Standing, broad learning, and application claims outside the measured receivers remain open.

7. THEORY NETWORK AND PRESENTATION PROTOCOL

CONTENTS OF THIS PART

Dependency register	Presentation protocol
Claim receipt	Mathematical notation
Figures	Source and citation
Reader notes	Typst package policy
Journal architecture	Source ledger for this volume
Present boundary	

The value of a foundation volume is not the number of analogies it can hold. It is whether a reader can identify which definitions support which derivations, measurements, and open questions. The following register is the initial theory network. Node identifiers are editorial addresses, not new machine state.

7.1. Dependency register.

ID	Relation	Depends on	Strongest current support
D1	actual occurrence	-	working definition
D2	oriented incidence and $\partial^2 = 0$	D1	definition + exact proposition
D3	atomic event	D1	direct constraint + implementation
D4	RIDE / FOUND / OPEN	D2, D3	definition + bounded measurement
D5	relative completion	D2, D3	exact proposition + implementation
D6	receiver-exact compression	D3, D5	linear theorem + bounded implementation
D7	learning as changed later conduct	D3, D4	definition + matched measurements
F1	formal exponential generator	D1	exact formal derivation
F2	Euler prime-power exponential	F1	exact formal/convergent theorem
F3	homogeneous aperture recurrence	F1	exact theorem + measurement
F4	theta involution	D4, F1	exact finite derivation + measurement
F5	aperture-tail balance	F4	inherited Poisson law + measurement
R1	completed RH positivity carrier	F2, F4, F5	OPEN

TABLE 11. Foundation and formula nodes in the initial dependency register.

ID	Relation	Depends on	Strongest current support
C1	hypergraph/rewrite specialization	D1-D5	direct correspondence
C2	manifold/coarea specialization	D2, D5	external theorem + correspondence
C3	annular/fractal return	D4, F1	local exact match + bounded analogy
C4	Ricci/physics specialization	D4, C2	external theory + typed bridge
L1	neural realization versus relation	D4, D6	derivation + ML comparison
L2	source-departed inherited law	D3, D7	exact measurement
L3	cultivation and local completion	L1, L2	exact bounded measurement
M1	machine-owned successor	D3	source + test evidence
M2	world consequence and rest	M1	host/CUDA measurement
M3	outgoing physical factor	D5, D6, M1	bounded host measurement

TABLE 12. Comparative, learning, and machine nodes. A later paper may refine a node but must not silently change the meaning of its dependencies.

The graph prevents three recurrent mistakes. First, a measured special case cannot become the definition on which it depends. Second, a comparison cannot promote itself into an exact cross-domain identity. Third, an open global bridge cannot borrow closure from several exact local lemmas whose missing composition has not been proved.

7.2. Presentation protocol.

Every substantive section begins by fixing four things in ordinary prose:

1. the exact question;
2. why standing evidence cannot already answer it;
3. the smallest qualitative observable or artifact sought; and
4. the stopping condition.

The manuscript then uses ordinary mathematical exposition. Provenance boxes are reserved for places where authorship, source route, or an open boundary would otherwise be ambiguous. The page does not become a dashboard.

The front outline exposes headings through level three. Every major journal part begins with a compact local map of its named subsections. Both are hyperlinks in the digital artifact; neither duplicates claims or evidence. The global outline answers “where is this relation?” while the local map answers “what distinctions organize this part?”

7.2.1. Claim receipt.

A load-bearing claim must let a reader recover:

(author, body, Current, Standing, source, receiver),

(frame, grain, grade, support, foil, open boundary).

These fields may be distributed across prose, theorem, caption, and source note. They should not be repeated ceremonially. The first substantial use of a measurement carries the complete receipt; later uses link back.

7.2.2. *Mathematical notation.*

Exact analytical structure uses integers, rational numbers, algebraic values, formal series, or declared analytic limits. Floating-point geometry may render testimony but cannot become causal evidence. The five judgments in J1 remain visually distinct. A symbol acquires a receiver, frame, grain, aperture, or source index whenever changing that index can change the result.

Theorem-family labels are bold and their bodies are roman. Italic is reserved for a term being defined, ordinary emphasis, variables, and the proof label; it does not wash across an entire definition, lemma, corollary, proposition, axiom, theorem, or open problem.

RIDE, FOUND, and OPEN are reserved words. “Residual” names either the complete path pair or a projection in a declared chart. “Holon,” “manifold,” “field,” “expert,” “probability,” “loss,” and “memory” never carry an unindexed universal meaning.

7.2.3. *Figures.*

Every figure caption states what kind of figure it is:

- **formal:** exact consequence of written definitions;
- **causal:** actual event incidence or measured lifecycle;
- **observer:** projection or testimony from exact material;
- **comparison:** an explicit map between domains; or
- **hypothesis:** a proposed, unmeasured relation.

The relational vector grammar uses four primitives: nodes for supplied occurrences, lines for declared incidence or transport, polygonal faces for one captioned geometric role, and dashed lines for a named receiver quotient, world passage, proposed bridge, or OPEN comparison. Position, proximity, size, and color do not create identity, chronology, or incidence. Actual order requires an arrow and a source chronology. Actual sharing is rendered as one shared occurrence; equal disconnected faces remain plural. A triangle names whether it is boundary, comparison, rewrite hinge, abstract simplex, or geometric realization. Accessibility is redundant: labels and line styles carry every distinction made visually.

7.2.4. *Source and citation.*

Primary papers, canonical texts, official documentation, and exact laboratory artifacts carry theorem, architecture, and measurement claims. Visual explainers such as 3Blue1Brown are cited for explanatory lineage and intuition. Wolfram Physics supplies an explicit comparison target for hypergraph rewriting and event causality; it does not make Eros a Wolfram model. A mutable web source records version or access date when it bears a substantive claim.

Internal records are routed through the current `src/soma/README.md` relation. Dated research deposits preserve derivation and authorship; they are not presumed

current merely because they are newer. A superseded interpretation may be cited as provenance only alongside its correction.

7.2.5. *Reader notes.*

Typst // comments are private editorial notes and do not render. A visible reader note carries an author and a disposition: `open`, `answered`, `incorporated`, `rejected`, or `parked`. Incorporation moves the idea into prose with its proper provenance. It does not turn a provisional question into a theorem.

READER NOTE - BRANDON (OPEN). This space is intentionally available for objections, alternate derivations, missing sources, and requests to split or merge a theory node. A note remains visibly yours until you decide its disposition.

7.3. Typst package policy.

The page grammar is owned by `unequivocal-ams` 0.1.2: US-letter AMS margins, New Computer Modern, numbered sections, and theorem/proof treatment. The local library does not override its page or heading system. It supplies the journal’s roman theorem-family body, global/local navigation, repeated holonic notation, restrained provenance notes, and Fletcher diagrams.

Package	Policy	Jurisdiction
<code>unequivocal-ams</code> 0.1.2	foundation	page, theorem, proof, bibliography
<code>fletcher</code> 0.5.8	active	commuting diagrams, incidence, dependency; wrapped locally
<code>quill</code> 0.7.3	conditional	only specified qubits, gates, wires, and measurements
<code>typed-dsa</code> 0.3.1	conditional	declared algorithms and external data structures
<code>typed-smiles</code> 0.8.0	conditional	source-declared chemical structures and reactions

TABLE 13. Package policy. Optional packages enter only when a real paper uses their native semantics.

`Quill` is not a generic metaphor for quantum relation. `typed-dsa` may render an exact data structure but cannot infer causal incidence. `typed-smiles` supplies chemistry-world technology; a SMILES string or rendered molecule is not Soma ontology. A local wrapper is added only after two or more manuscripts repeat a semantic need.

7.4. Journal architecture.

The five streams remain topical receivers:

Stream	Core question	First paper family
J1	What is the minimal calculus?	Occurrence, event, comparison, completion, compression
J4	How do finite formula faces grow?	Exponential/Euler apertures and theta boundary
J5	Which cross-domain maps survive?	Typed geometry, physics, life, and culture atlas
J3	What counts as learning?	Inherited ecology, retriangulation, and parent-on-OPEN
J2	What did the machine establish?	Atomic owner, world return, rest, and physical factor

TABLE 14. Standing journal streams in the requested development order. They are not a work queue.

Future entries should use:

```
entries/YYYY-MM-DD_slug/
  entry.typ
  figures.typ
  notes.typ
```

```
papers/short-title/
  main.typ
  sections/
  figures/
  references.bib
```

An entry is authored once and may be referenced from several streams. The source directory, Formula, research deposits, and observation artifacts remain the claim substrate. **PAPERS/** is a reading and publication surface, not a second owner or scheduler.

7.5. Source ledger for this volume.

Code	Use	Decisive internal route
S1	claim routing	README
S2	foundational mechanics	Elementary Mechanics
S3	historical/formal spine	Formula
S4	mathematical analogy catalog	Theory Map
S5	formula generator	Infinite Formula
S6	aperture recurrence	Aperture Grows
S7	rewrite/triangle derivation	Hypergraph / Triangle
S8	scale-turn derivation	Annulus
S9	hyperarea/receiver derivation	Hyperarea
S10	Ricci/physics derivation	Ricci Trace
S11	ML interpretation correction	Network Is One Realization
S12	machine owner correction	Event Emanates Successor
O	bounded measurements	Observation RESULTS cited in place

TABLE 15. Internal source ledger. Descriptive links at first use point to the complete paths; abbreviations here keep the table readable.

7.6. Present boundary.

This foundation is intentionally nonfinal. It now supplies what the first prospectus lacked: typed definitions, reusable derivations, exact evidence cuts, cross-domain non-equivalences, and a dependency network. Its immediate open work is scholarly rather than rhetorical:

- formal semantics and countermodels for the calculus;
- arbitrary-degree covariance for formula apertures;
- the completed RH trace and positivity carrier;
- boundary-preserving neural retriangulation with physical cost;
- grounded later consequence outside retained transformer faces;
- general confluence and device-resident cellular execution; and
- domain-specific physical predictions with units and falsifying controls.

None is scheduled by appearing in this list. Each becomes active only when a present question selects its README evidence route and fixes a new stopping condition.

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